Main Figures

Parallel transcriptomics reveals insect developmental shifts and gall-associated host transcriptional differences in *Quadrastichus erythrinae* and *Erythrina variegata*

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Figures 1–6

Figure 1. Study system and experimental design

Figure 2. Insect GO and KEGG enrichment

Figure 3. Insect secretome and candidate families

Figure 4. Plant GO and KEGG enrichment

Figure 5. Plant-defense candidates

Figure 6. MEROPS inhibitors and peptidases

Figure 1

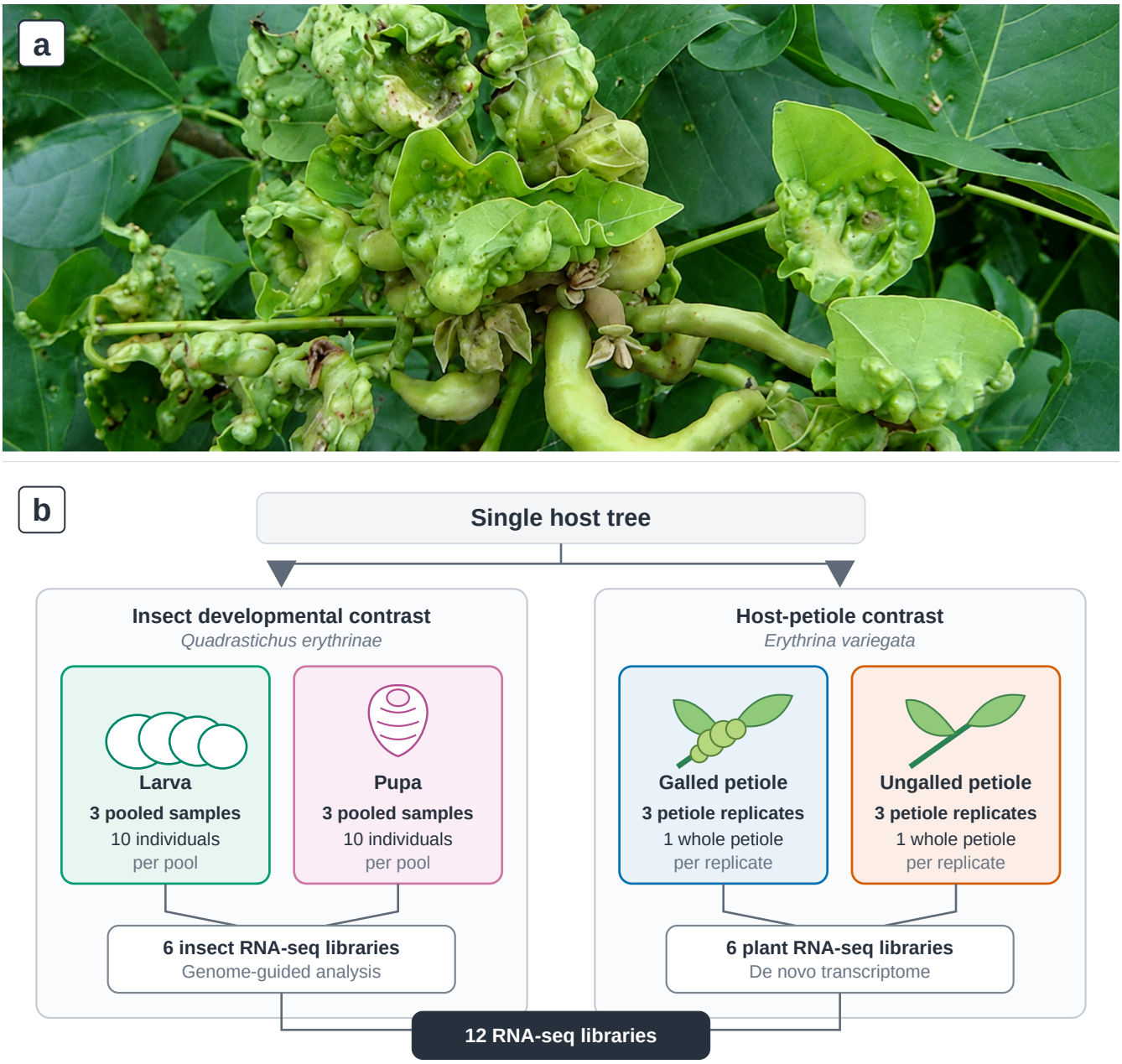


Fig. 1. Study system and experimental design. (a) *E. variegata* galls induced by *Q. erythrinae*. Image cropped from Lin et al. [9], Fig. 1B (CC BY 4.0). (b) Separate larva–pupa and galled–ungalled petiole comparisons from a single tree sampled once in Taipei, Taiwan, in November 2020. Each insect stage comprised three pools of 10 individuals from different galled petioles; each plant condition comprised three whole-petiole replicates (GP, galled; UP, ungalled).

Figure 2

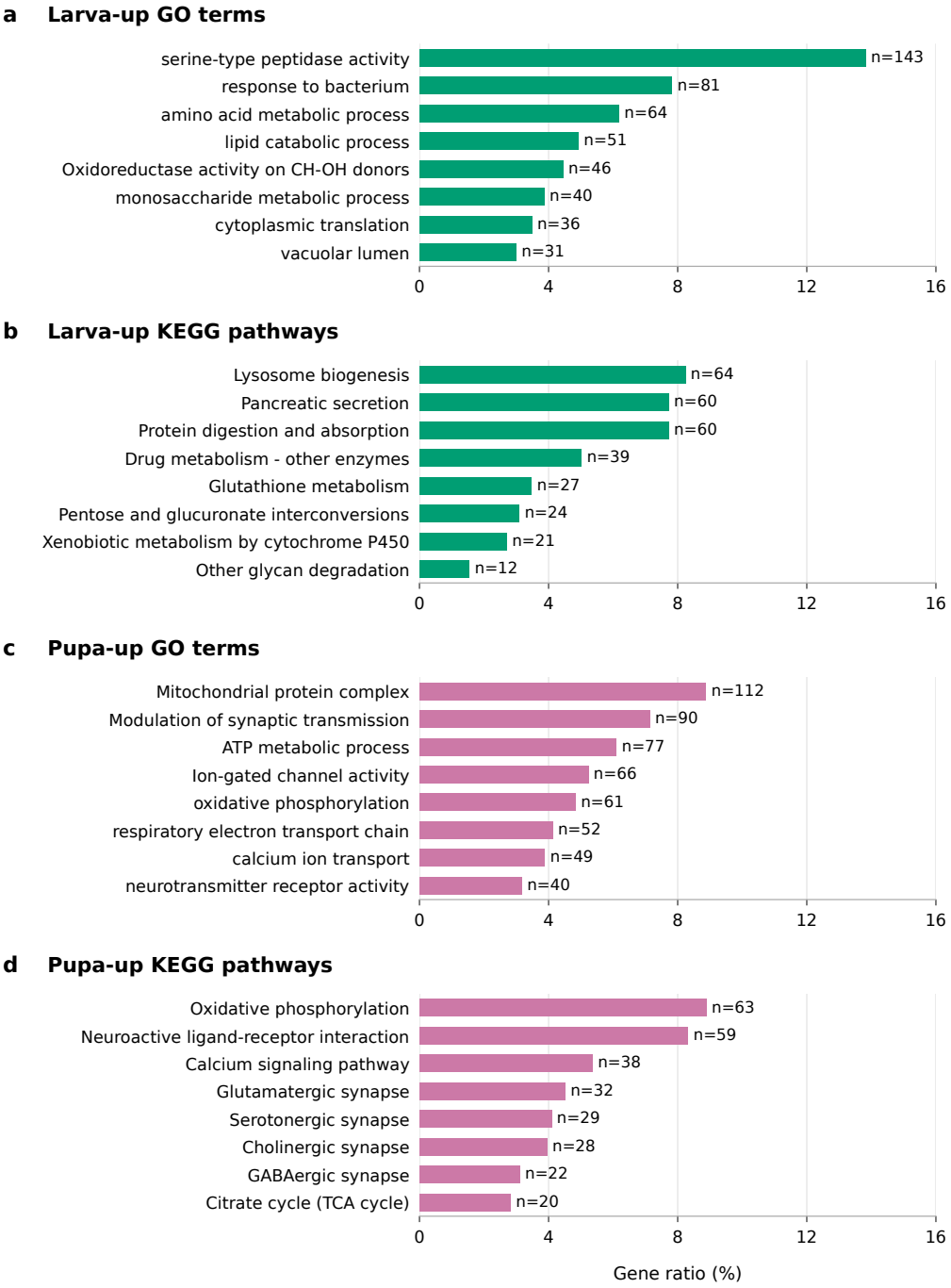


Fig. 2. Representative GO terms and KEGG pathways enriched among larva-up and pupa-up genes in *Q. erythrinae*. (a) Larva-up GO terms. (b) Larva-up KEGG pathways. (c) Pupa-up GO terms. (d) Pupa-up KEGG pathways. Larval and pupal categories are shown in green and pink, respectively. Bars show the percentage of GO- or KEGG-annotated genes in each directional gene set assigned to the indicated category; labels show gene counts. All displayed categories had BH-adjusted $P < 0.05$. Representative categories were selected to reduce redundancy, although categories may share contributing genes. Complete results are provided in Supplementary Tables S4 and S5.

Figure 3

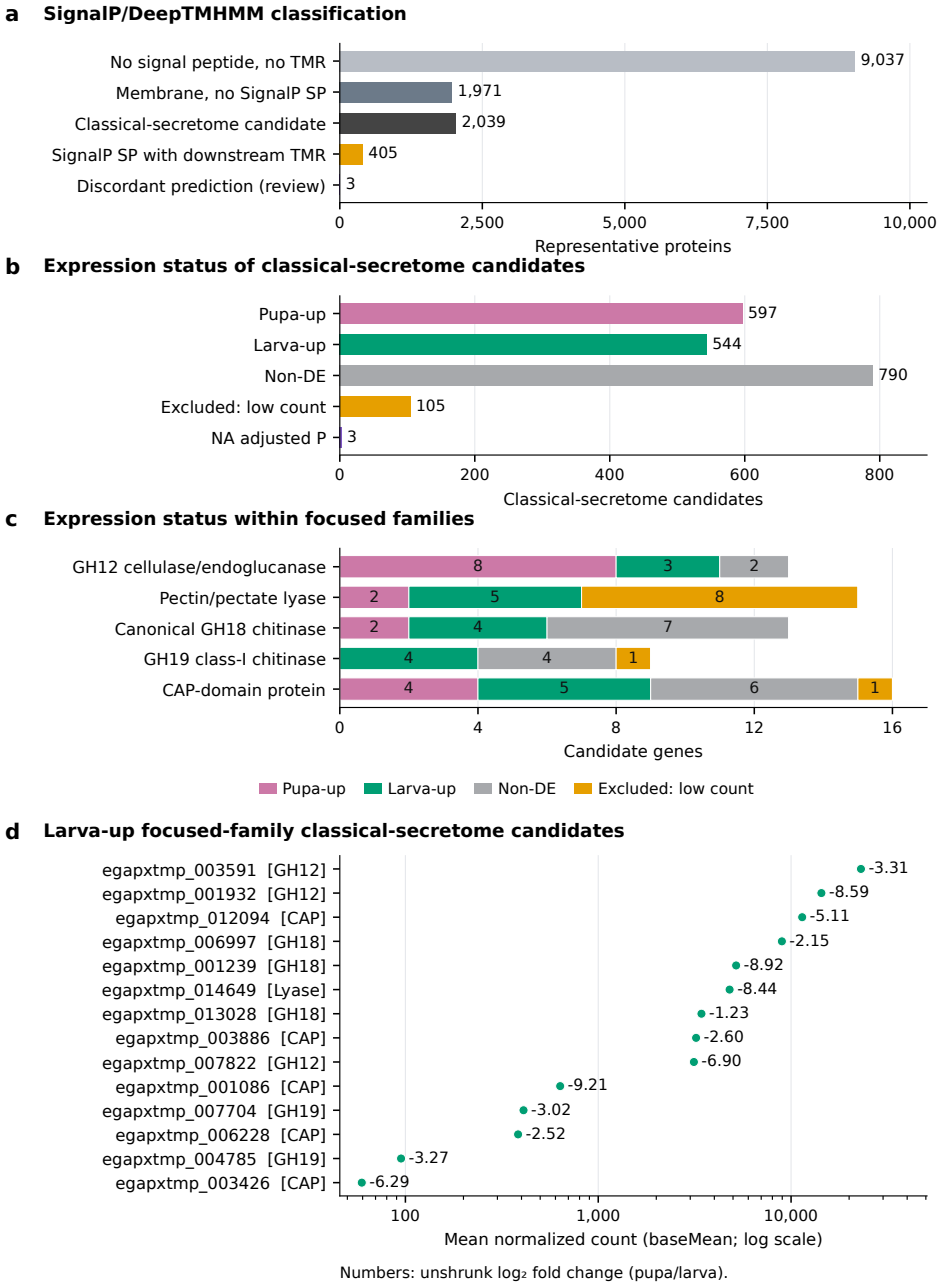


Fig. 3. Predicted classical-secretome and focused candidate-family profiles in *Q. erythrinae*. (a) Classification of 13,455 representative proteins using SignalP and DeepTMHMM. (b) Expression status of the 2,039 proteins meeting the conservative classical-secretome criterion: a predicted SignalP signal peptide and no predicted DeepTMHMM transmembrane region. (c) Expression status within five focused candidate-family inventories. (d) Mean normalized counts for the 14 focused-family genes that were both larva-up and classical-secretome candidates. In (b) and (c), green and pink indicate larva-up and pupa-up genes, respectively. Points in (d) are uniformly green; bracketed abbreviations identify candidate families, and numerical labels show unshrunk $\log_2$ fold changes for pupa versus larva. Differential expression required adjusted $P < 0.01$ and $|\log_2$ fold change$| \geq 1$. Non-DE genes did not meet both criteria; low-count exclusions and unavailable adjusted P values are shown separately. Classical-secretome classification predicts entry into the classical secretory pathway; extracellular localization and effector activity remain to be tested.

Figure 4

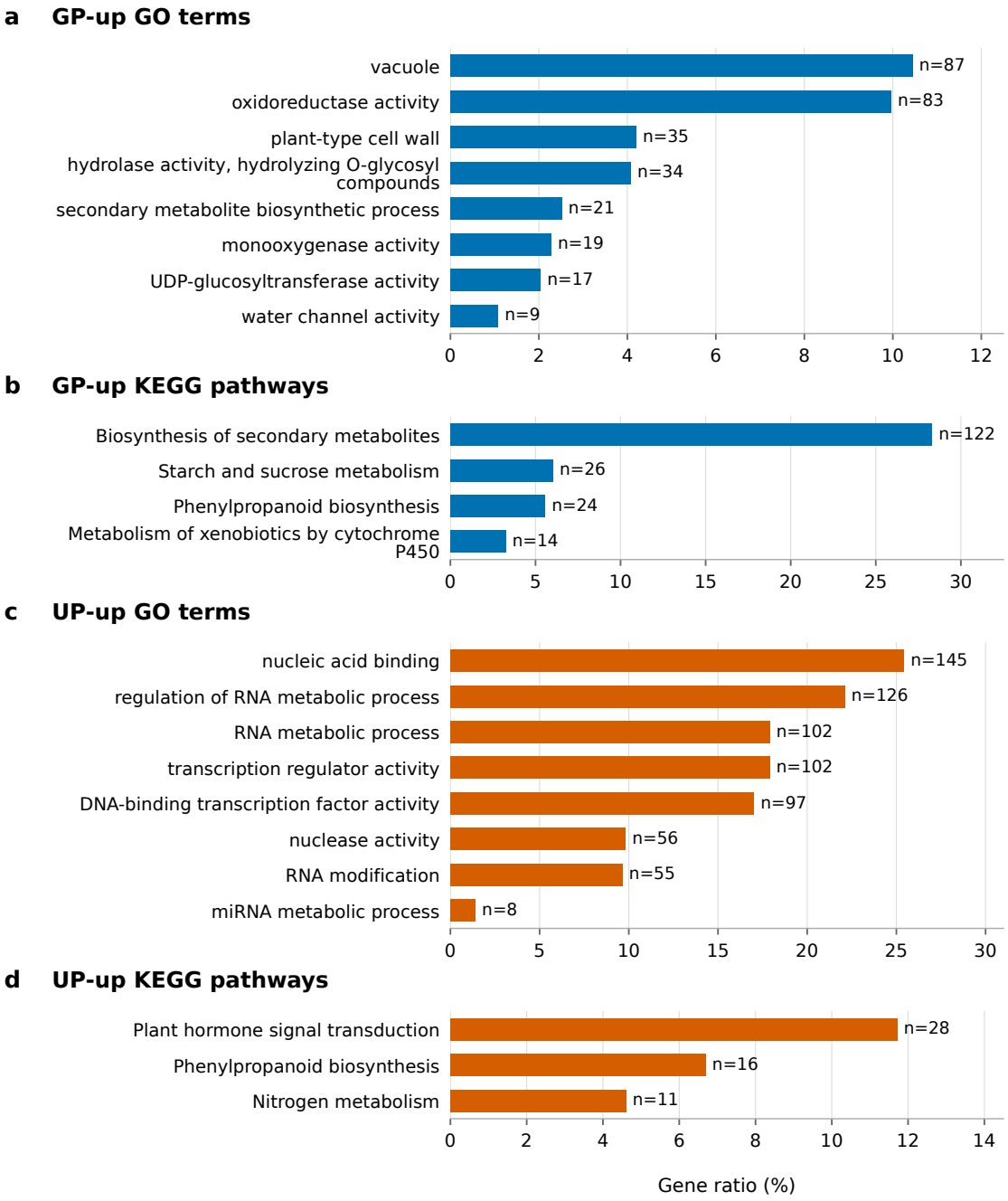


Fig. 4. Representative functional enrichment associated with differential expression in galled and ungalled *E. variegata* petioles. Currently resolved GO terms and significant KEGG pathways are shown for GP-up genes (a, b; blue) and UP-up genes (c, d; orange). Bars indicate the percentage of annotation-eligible genes in each directional set assigned to a category, and labels show gene counts. All displayed categories had BH-adjusted $P < 0.05$. Redundant GO terms were omitted from the figure but retained in Supplementary Table S11. The broad KEGG category metabolic pathways and the category chemical carcinogenesis-DNA adducts were omitted from panel b but retained in Supplementary Table S12. The latter shares all 14 GP-up contributing genes with metabolism of xenobiotics by cytochrome P450 and therefore does not provide an independent gene-level enrichment signal.

Figure 5

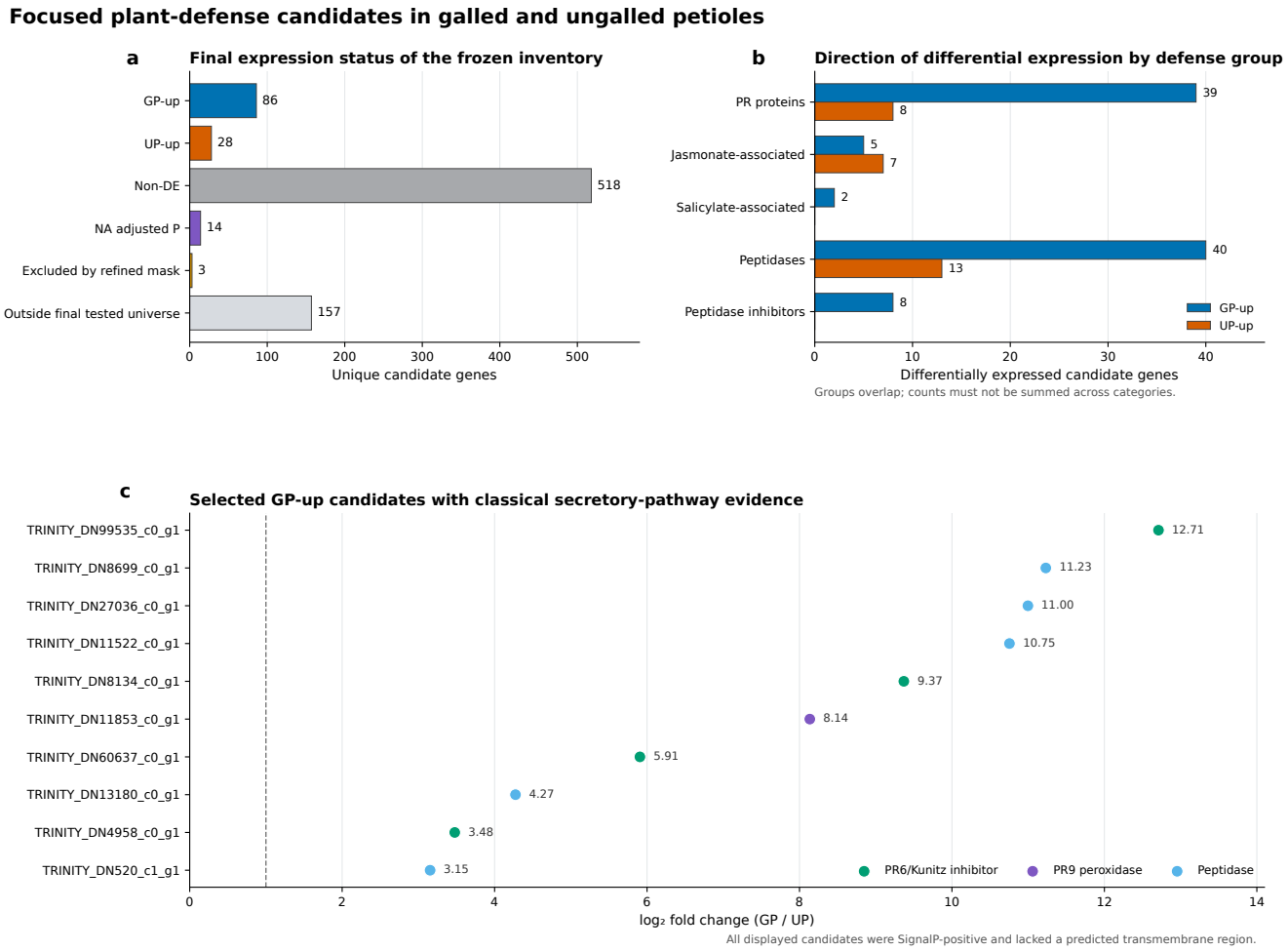


Fig. 5. Expression patterns within the frozen plant-defense candidate inventory. (a) Final refined-mask expression status of 806 unique candidate genes. (b) Numbers of GP-up and UP-up genes within five overlapping defense groups. Group-specific counts are not additive because some genes belong to multiple groups. (c) Selected GP-up candidates that were SignalP-positive and lacked a predicted transmembrane region. Points are colored by candidate class, and labels show $\log_2$ fold change for GP versus UP. The dashed line marks $\log_2$ fold change = 1. SignalP-positive/no-transmembrane-region classification indicates predicted entry into the classical secretory pathway, not confirmed extracellular localization or biological activity.

Figure 6

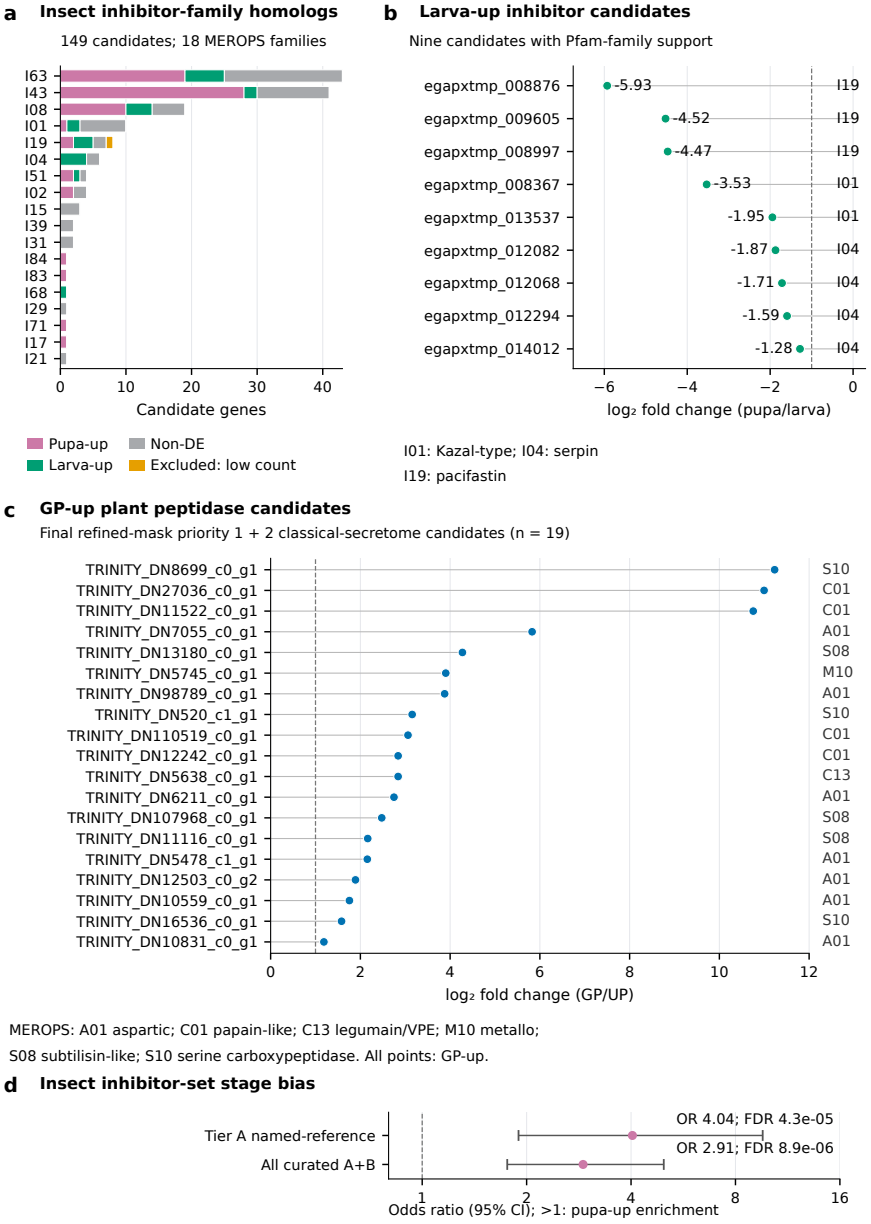


Fig. 6. Stage-associated insect inhibitor-family homologs and GP-up plant peptidase candidates. (a) Expression-status distribution of 149 curated classical-secretome insect inhibitor-family homologs across 18 MEROPS families. (b) Nine larva-up candidates with the expected Pfam-family support. Negative $\log_2$ fold changes represent higher larval transcript abundance in the pupa-versus-larva contrast. (c) Nineteen priority plant peptidase candidates retained as GP-up after the final refined-mask DESeq2 analysis. Positive $\log_2$ fold changes represent higher GP transcript abundance in the GP-versus-UP contrast. Candidates in (b) and (c) are colored green and blue, respectively; adjacent codes identify MEROPS families. Stems in these panels connect estimates to zero and are not confidence intervals. (d) Pupa-up representation of the complete curated insect inhibitor set and the Tier-A named-reference subset. Points show odds ratios and horizontal lines show 95% confidence intervals. Dashed lines mark $\log_2$ fold changes of −1 in (b) and +1 in (c), and an odds ratio of 1 in (d). Non-DE genes did not meet adjusted $P < 0.01$ and $|\log_2$ fold change$| \geq 1$; low-count exclusions and unavailable adjusted P values are separate categories. GP, galled petiole; UP, ungalled petiole. These homology-based profiles identify candidates for biochemical testing.